

NTC/2063/xx

DVB

Demodulator



TO WHOM IT MAY CONCERN**EC DECLARATION OF CONFORMITY**

We,

NEWTEC CY nv.

declare that our product

DVB+8+16 Satellite Demodulator NTC/2063/xx

to which this declaration relates is in conformity with :

the requirements of the EMC Directive 89/336/EEC (1)
the requirements of the Low Voltage Directive 73/23/EEC (2)
the requirements of the CE Marking Directive 96/68/EEC

in accordance with the relative standards listed below :

(1) : EN 50081-1:92
Electromagnetic Compatibility - Generic emission standard
part 1 : residential, commercial and light industrial environment

EN 50082-1:92
Electromagnetic Compatibility - Generic immunity standard
part 1 : residential, commercial and light industrial environment

(2) : EN 60950:92 incl. A1-A4
Safety of information technology equipment

Done at Antwerp, on February 27, 1998

Dirk Breynaert,
Managing Director

Relevant EMC information (to FCC rules)

This equipment has been tested and found to comply with the limits for a class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

SAFETY

Please read this chapter before installation and use of the instrument

This equipment is provided with a protective earthing ground incorporated in the power cord. The mains plug shall only be inserted in a socket outlet provided with a protective earth contact. Any interruption of the protective conductor, inside or outside the instrument, is likely to make the instrument dangerous. Intentional interruption is prohibited.

The instrument described in this manual is designed to be used by properly-trained personnel only. Adjustment, maintenance and repair of the exposed equipment shall be carried out only by qualified personnel who are aware of hazards involved.

No operator serviceable parts inside. Refer servicing to qualified personnel. To prevent electrical shock, do not remove covers.

For the correct and safe use of the instrument, it is essential that both operating and servicing personnel follow generally accepted safety procedures in addition to the safety precautions specified in this manual. Warning and caution statements and/or symbols are marked on the instrument when necessary.

Whenever it is likely that safety protection is impaired, the instrument must be made in-operative and secured against unintended operation. The appropriate servicing authority must be informed. For example, safety is likely to be impaired if the instrument fails to perform the intended measurements or shows visible damage.

Rack mounting instructions :

When mounted in a standard 19 inch equipment rack, the device must be sustained by L-profiles. Fixing the device with 4 front panel screws only will damage it and could result in injury.

WARNINGS

- Do not use the equipment in damp surroundings. Avoid direct contact with water.
- Never place the equipment in direct sunlight.
- Never cover the fan opening when the equipment is in use as this will impede the ventilation required.
- The outside of the equipment may be cleaned using a lightly dampened cloth. Do not use any cleaning liquids containing alcohol, methylated spirit or ammonia etc.
- For continued protection against fire hazard, replace line fuses only with same type and rating (5 x 20 mm, 3.15 A 250 V type T or slow-blow).

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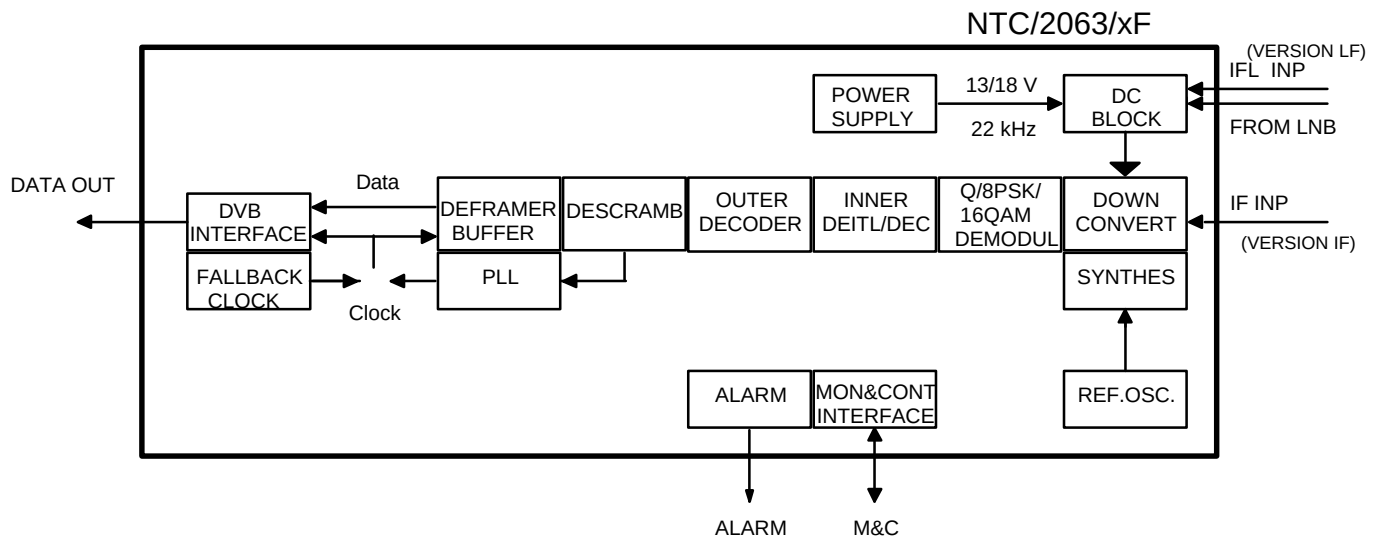
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1. TECHNICAL DESCRIPTION

1.1. Block Diagram



1.2. Software History

NTC/6094/AA Main Controller Software

v 1.00	23 Feb 1998	first release for production
v 1.02	31 Mar 1998	adapted for NTC/3362/xG HSSI interface module
v 1.10	27 May 1998	changed interface with demodulator DSP, needs NTC/6108/AA v1.1x or later
v 1.11	25 Jun 1998	adapted for NTC/2063/xN – 1.42 to 22.5 Mbaud Demodulator
v 1.20	24 Sep 1998	adapted for DVB modulation standard (former DVB standard renamed to NTC standard)
v1.30	9 Feb 1999	adapted for IF interface (NTC/2062/lx)
		adaptation for NTC/2063/lx demod with Newtec converter (LO 1000 MHz)
v1.40	12 Apr 1999	addition of interface NTC/3389/HA
v1.50	06 Sept 2000	adapted for interface board NTC/3362/HE and NTC/3362/DE extended HSSI

NTC/6108/AA Modulator DSP Software

v 1.00	5 Feb 1998	first release for production
v 1.10	27 May 1998	changed interface with main controller, needs NTC/6094/AA v1.1x or later

2. Installation

Warning : This equipment is provided with a protective earthing ground incorporated in the power cord. The mains plug shall only be inserted in a socket outlet provided with a protective earth contact. Any interruption of the protective conductor, inside or outside the instrument, is likely to make the instrument dangerous. Intentional interruption is prohibited. The type of mains plugged shipped with each instrument depends on the country of origin.

Caution : Do not connect AC power until you have verified that the line voltage is correct and the proper fuses are installed. Damage to the equipment could result. The equipment requires a power source of either 50/60 Hz at 100 V or 50/60 Hz at 240 V. The voltage ranges for these nominal voltage values are shown in the table :

POWER REQUIREMENT	
Input voltage	90 - 130 V _{rms} or 180 - 260 V _{rms} auto select
Frequency	47 to 63 Hz
Power	150 VA max. / 70 VA nom.
Current	100 V/0.68 A or 240 V/0.28 A nominal

Caution : This instrument has autoranging line voltage input ; be sure the supply voltage is within the specified range.

Checking the fuses : The recommended fuses are size 5 by 20 mm, rate T 3.15 A, 250 V (UL and IEC approved).

The line fuses are housed in a small container above the rearpanel power connector. To check the fuses, insert the tip of a screwdriver in the slot at the middle of the container and pry gently to extend the container. The fuse container is attached to the line module; it cannot be removed.

Remark : The power entry module has a marking of 6.3 A max. for the fuse on the fuse body. This value is applicable to the power entry module only, not to the instrument. The recommended fuse ratings are printed on the rearpanel nearby the fuse holder.

2.1. Inspection and test

Remove the unit from the box in which it is shipped. Keep the box for later transport of the unit.

Verify the unit for damage. If everything seems all right, power can be applied to the unit. Mains socket and fuses are located at the rearpanel of the unit.

When powered, verify that the display gives following message :

NEWTEC Cy HR DVB+8 Demodulator (version - date)
system test passed display : ↑ ↓ ← →

If there is no message at all (display not lighted) or if SYSTEM TEST FAILED is displayed, the unit is defective and has to be returned for servicing.

2.2. Functional self test

No specific functional self test is needed.

The unit has been factory tested according to acceptance test procedures and is ready for operation.

When also a modulator, example NTC/2062/xx, is available, the next test can be performed.

Take care to disable the DC at the modulator output and/or the demodulator input or place a DC-block in the connection between them. If one of the devices has DC on its I/O, then both devices can be damaged

- connect the L-band output of the modulator to the L-band input of the demodulator,
- connect the data output of the demodulator to the data input of the modulator,
- set the output level of the modulator to -35.0 dBm, and use additional attenuators if the receive level is still too high (level saturation alarm on demodulator)
- set the deframer buffer output clock (or the interface clock source) of the demodulator to fallback or internal clock
- set the transmit data to internal PRBS, the transmit clock to internal and the interface MPEG framing to internal for the modulator,
- take care that carrier frequency, symbol rate, modulation, FEC and RS coding, framing and interface match for modulator and demodulator.

Verify that there are no active alarms at the modulator and that the demodulator correct receives the PRBS pattern. Thus no alarms and errors for some time.

2.3. Mounting

If the functional self test is OK, then the unit can be installed at the required location.

The environment should be relatively dust free, free of excessive vibration and the ambient temperature below 40°C.

Air intake for cooling is via holes at the side of the unit and the fans at the back panel. Air flow should not be obstructed. Therefore, the unit has to be placed on a flat surface, leaving some space at the sides of the unit.

Warning : When mounted in a standard 19 inch equipment rack, the device must be sustained by L-profiles. Fixing the device with 4 front panel screws only will damage it and could result in inquiry.

When in operation, the internal temperature should not exceed the limit of 70°C.

2.4. Recommended default settings

The following values for the system and control parameters allow normal operation of the device.

The parameters marked with TBD depend on the target link or the preference of the user.
The indicated default values are the values saved in permanent memory for new devices.

2.4.1. System parameters

The system parameters are to be adapted at installation. These parameters inform the device about the surrounding hardware and how to use it.

All these settings are maintained in permanent memory and saved when changed. After power-on the device uses the settings as were before power-down.

test functions	: disabled	
alarm mask	: 0000 hex	
device address	: TBD	default 51
device control	: local	
LNB power supply A	: disabled	
LNB power supply B	: disabled	

Remark that the device always starts with the test functions disabled and local device control.

The device address for remote M&C must be changed so that it is unique in the target group for the remote control unit.

Remark that the default remote interface is RS485. This selection is done via straps inside the device (see 7.5.). The selection of the RS232 interface is indicated on the back panel when done by the factory (on request).

2.4.2. Control parameters

The control parameters define the actual operation status of the device. They can be changed by the operator at any time.

All these settings are saved in permanent memory as a configuration. They are saved and read under operator control via dedicated system functions. After power-on or reset, the device uses the settings from configuration 0.

At installation the default control settings in configuration 0 must be adapted in function of the target link. When needed, more target links can be defined and saved as configurations 1 to 4.

interface rate	: TBD	default : 8.294118 MHz
symbol rate	: TBD	default : 6.000000 Mbaud
FEC-rate and modulation	: TBD	default : QPSK, FEC 3/4
LNB A control	: not applicable (fixed)	
LNB B control	: not applicable (fixed)	
LNB input selection	: input A	
excess bandwidth	: auto	
IF(L) frequency	: TBD	default : 1.170 GHz
acquisition range	: 1000 kHz	
spectrum inversion	: auto	
descrambler	: on	
MPEG framing	: external, 188 byte frames	
average buffer size	: 32 MPEG frames	
deframer buffer output clock	: satellite	
sub-D interface	: DVB-SPI-LVDS	
coaxial interface	: off	
modulation standard	: DVB	

2.4.3. Operational configuration : DVB standard

The DVB standard specifies MPEG sources, with co-directional clock from demodulator to decoder.

- MPEG framing : external
- buffer output clock : satellite (slaved to received clock)

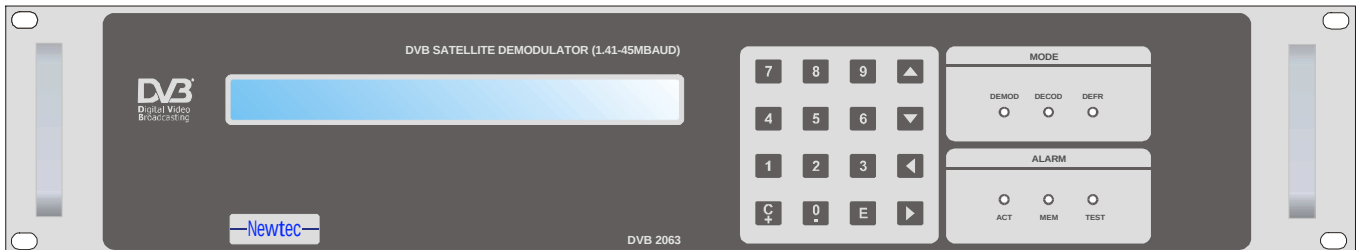
2.4.3. Operational configuration : Non-MPEG Source

In the case that a non-MPEG source is used, then the modulator can insert MPEG sync words to the data. This allows coding synchronization in the demodulator. The demodulator can then strip off this MPEG sync words, so that input data only remains. Remark that on the modulator the external transmit clock is required, to prevent data stuffing.

So :

- MPEG framing : internal
- buffer output clock : satellite (slaved to received clock)

3. FRONTPANEL



3.1. LCD

The 2x40 Liquid Crystal Display (LCD) provides the operator with all the required information about the selected parameter.

The top line of the LCD gives the selected parameter and its present value. The bottom line indicates the menu to which the parameter belongs and how it can be manipulated.

The parameters are grouped in four menus :

- the **system menu**, which gives the general device management parameters and the hardware related parameters
- the **control menu**, with the main operation control functions
- the **alarm menu**, which informs about device critical functions
- the **monitor menu**, with operation related variables measured by the device.

When the test functions are enabled, then a fifth menu becomes available to the operator. This *test menu* contains functions which are beyond the normal operation of the device.

Remark that the instructions for manipulating a parameter are only given when the operator is allowed to change this parameter via the keypad. If no instructions are given then the parameter can not be handled for one of the following reasons :

- the parameter is monitoring only (monitor menu)
- the parameter may only be changed during test or installation
- the device operates under remote control
- the value of the parameter is forced by the value of other parameter, in the case of coupled parameters.

The information on the LCD is updated four times per second.

3.2. Keypad



The 16 front panel keys allow the operator to select the information given on the LCD and to change parameters, if the device operates under local control.

The **<->-** and **<^>-**keys are used to select between the main menus of the device. Once in the desired menu, system, control, alarm or monitoring, then the needed sub-menu can be found with the **<↵>-** and **<®>-**keys.

On the bottom line of each submenu, it is indicated if and how the parameter can be manipulated by the operator.

In general, the **<E>-key**, which means “enter”, toggles between the possible selections. It also finishes the input of new numerical values and confirms critical changes.

The **<C>-key**, in the meaning of “clear”, empties the numerical input fields, without any action, and resets the memorized alarms and their event counters.

The digit keys **<0> up to <9>** are used for the input of numerical values and the selections of multiple options.

For the input of integer numbers, the last entered digit is interpreted as the units of the thus formed number. Remark that possible not given preceding digits are interpreted as “0”.

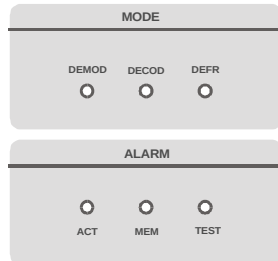
When the input of floating point numbers is needed, then the decimal dot is indicated on the LCD. The meaning of an entered digit is indicated by its location relative to the decimal dot. This includes that preceding “0”-s must be given. Remark that trailing digits, which are not given in, are set to “0”.

The **<+>-** and **<->-key** are used for the input of signed numbers and the tuning of some parameters.

Remark that for the input of signed numbers, the meaning of the **<C>-** or **<+>-key** and the **<0>-** or **<->-key** depend on when the key is pressed. If they are the first inputs of a number, then these keys are interpreted as sign. For the next inputs, they are treated as **<C>** or **<0>**.

When the **<+>-** and **<->-keys** are used for the tuning of a variable, then the step-size depends on the duration of the push. A small step is used for single pushes, while holding this keys for an extended time increases the step.

3.3. Indicator LEDs

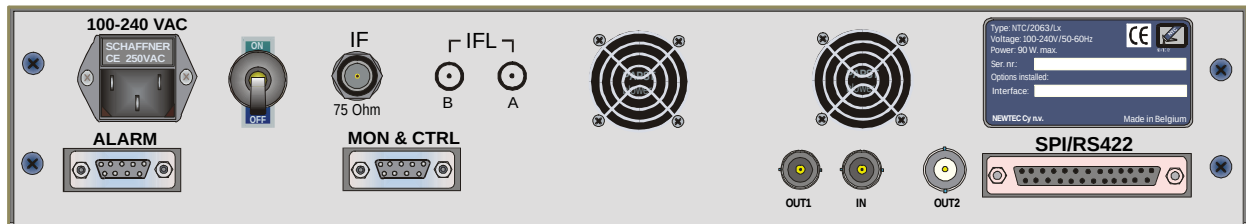


There are six LED-indicators :

- **demodulator mode** : the status of the demodulator
 - green : demodulator sync
 - off : carrier acquisition (demodulator unsync)
- **decoder mode** : the status of the decoder
 - green : the Viterbi decoder and the RS-decoder are synchronized
 - off : the Viterbi decoder and/or the RS-decoder are not synchronized
- **deframer mode** : the status of the unique word detector (MPEG framing)
 - green : the unique word detector is synchronized
 - off : no unique word synchronization
- **active alarm** : the combination of all the active demodulator alarms (see 5.3.)
 - red : the demodulator has an active alarm
 - off : the demodulator is all right
- **memorized alarm** : the combination of all the memorized demodulator alarms (see 5.3.)
 - red : an alarm has been detected and is no longer active (active alarm LED off) or is still active (active alarm LED on)
 - off : there has been no alarm since power-on or the last memorized alarm reset
- **test** : the demodulator operates in a non-standard mode
 - yellow : the test functions are enabled
 - off : the demodulator operates in the standard mode

The information on the LED's is updated four times per second.

4. BACKPANEL



4.1. Power supply

The power supply cable needs an IEC main socket at the side of the device.

The plug at the other side of the mains cable depend on the country of destination and has to be adapted at installation. Standard a cable with a European DIN 49441 two pin mains plug (Schuko) is delivered.

The line fuses are housed in a small container at the top of the power supply connector.

4.2. Mains On/Off Switch

The main on/off switch interrupts the power supply towards the device if OFF.

4.3. Fans

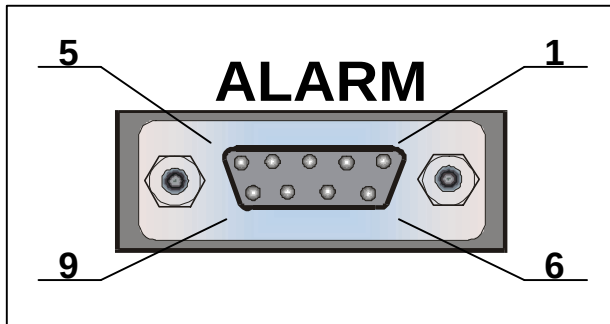
The fans provide the cooling airflow for the device. Take care not to obstruct these fans.

4.4. IF and IFL inputs

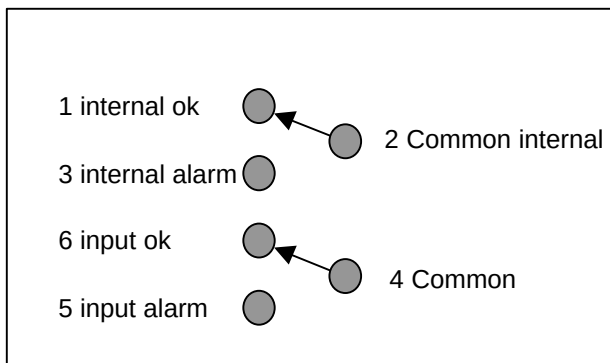
The two inputs for the NTC/2063/Fx are the L-band inputs IFL A and IFL B, while the NTC/2063/lx has one L-band input IFL A and one IF input.

The L-band inputs have a 75 Ω F connector. The IF input uses a 50 Ω BNC connector.

4.5. Alarm contacts



9-pin female sub-D connector
as seen on demodulator



1 - 2 - 3 monitoring output : demodulator input status (receiver ready), this is the summary (logical-OR) of the following alarms :

- demodulator sync (see 5.3.14.)
- Viterbi decoder sync (see 5.3.15.)
- unique word detection (see 5.3.16.)
- Reed-Solomon decoder sync (see 5.3.17.)
- deframer sync (see 5.3.18.)
- level saturation (see 5.3.10.)

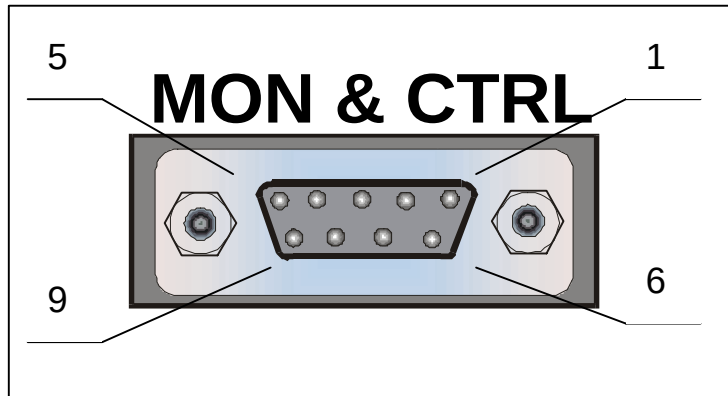
6 - 4 - 5 monitoring output : demodulator internal status, this is the summary (logical – OR) of the following alarms :

- +5 volt power supply (see 5.3.3.)
- -5 volt power supply (see 5.3.4.)
- +15 volt power supply (see 5.3.5.)
- -15 volt power supply (see 5.3.6.)
- device temperature (see 5.3.7.)
- synthesiser (see 5.3.8.)
- clock PLL (see 5.3.9.)
- buffer overflow (see 5.3.11.)
- buffer underflow (see 5.3.12.)
- data interface (see 5.3.13.)
- incompatibility indication (see 5.3.2.)

The outputs are always active.

During initialization or when the device is powered down, these contacts indicate alarm.

4.6. Serial Monitor and Control via RS485/RS232

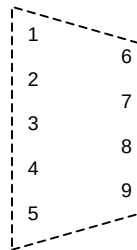


The modulator contains the hardware for the RS485 and RS232 interface. Selection between the RS-485 and RS-232 hardware is done via the straps near the serial M&C connector on the main board (see 7.5.).

9-pins female sub-D connector (inputs and outputs are related to this modulator).

RS485 serial interface:

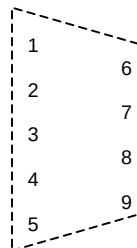
shield (ground)
N.C.
input : send data A
output : receive data A
signal ground



output : receive data B
N.C.
N.C.
input : send data B

RS232 serial interface:

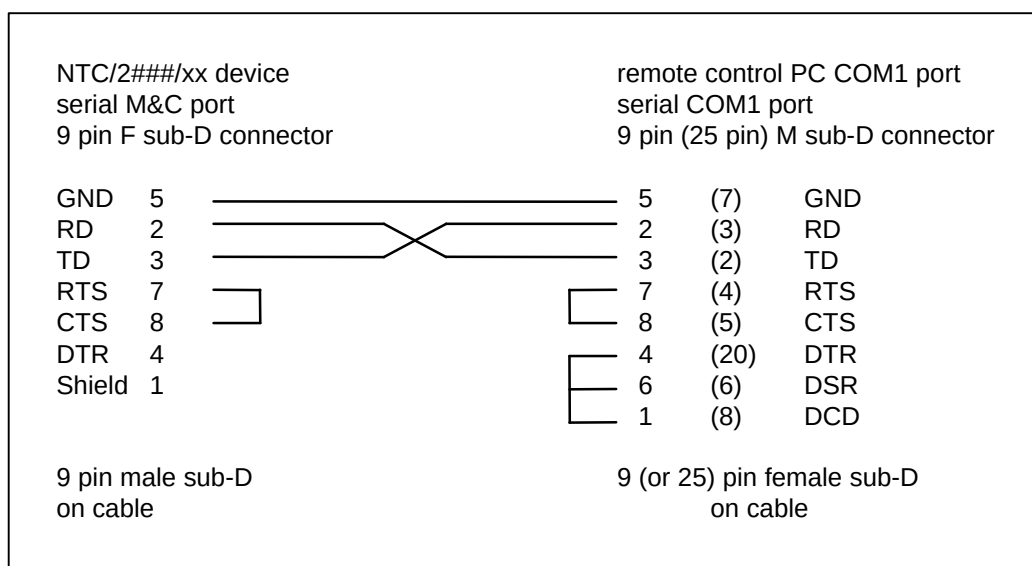
shield (ground)
input : receive data (RD)
output : transmit data (TD)
output : data terminal ready (DTR)
signal ground



N.C.
output : request to send (RTS)
input : clear to send (CTS)
N.C.

If the RS232 serial interface is used, the modulator behaves as a DTE, so the needed cable is a "null"-modem cable. Thus with the connections between "receive data" and "transmit data" crossed. The handshaking signals DTR and RTS are always on at the NTC/2###/xx device. Remote devices that need these signals can use them for serial communication. The device does not need CTS.

In most cases, the remote control device is a PC, where the serial COM1 port could be used. This port has a 9-pin or 25-pin male sub-D connector. For the typical layout of this connector (to be verified in the PC documentation), the following "null"-modem cable (option NTC/3267/AC) can be used:



WARNING! The use of the wrong hardware can damage this device, the remote control unit and possibly other devices on the bus.

4.7. Sub-D Interface

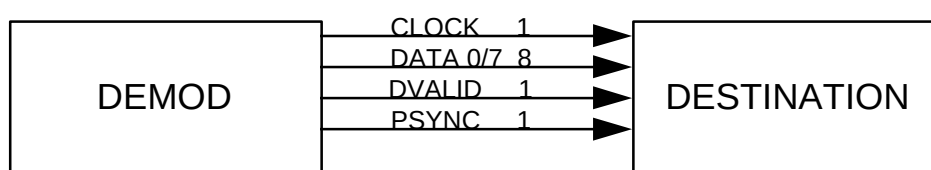
25-pins female sub-D connector : pin-out depends on the selected interface.

The type of cables and the connectors depend on the used interface module NTC/33xx/xx and are described in the data sheets of this module.

- for the DVB-SPI interface :

This interface requires the interface module NTC/3320/ZG, NTC/3326/xx or NTC/3327/xE.
The DVB-SPI interface is conform to EN 50083-9 "Interfaces for CATV/SMATV Headends and Similar Professional Equipment".

(*) signals and pinout :

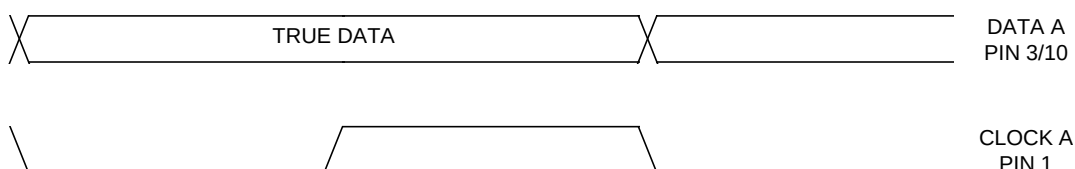


PIN	SIGNAL LINE	PIN	SIGNAL LINE
1	Clock A	14	Clock B
2	System GND	15	System GND
3	Data 7 A (MSB)	16	Data 7 B
4	Data 6 A	17	Data 6 B
5	Data 5 A	18	Data 5 B
6	Data 4 A	19	Data 4 B
7	Data 3 A	20	Data 3 B
8	Data 2 A	21	Data 2 B
9	Data 1 A	22	Data 1 B
10	Data 0 A	23	Data 0 B
11	DVALID A	24	DVALID B
12	PSYNC A	25	PSYNC B
13	Cable Shield		

Remark that the PSYNC signal is not required.

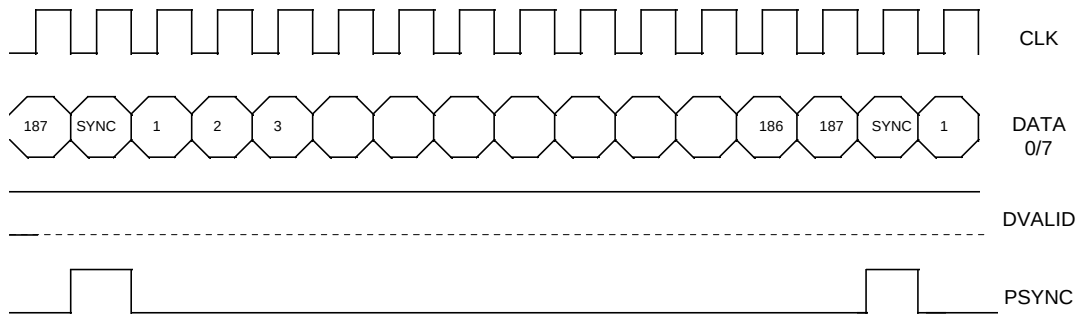
(*) electrical : V11 differential receiver
100 Ω termination
Accepts also balanced negative ECL levels as supplied by Philips equipment.

(*) timing : normal (as shown below) or inverted (test only)
programmable via keyboard or remote M&C I/F

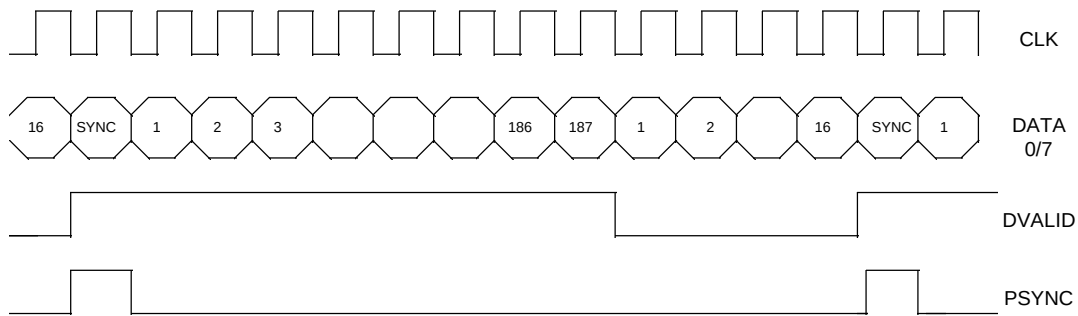


Normal Clock-Data relationship for SPI interface

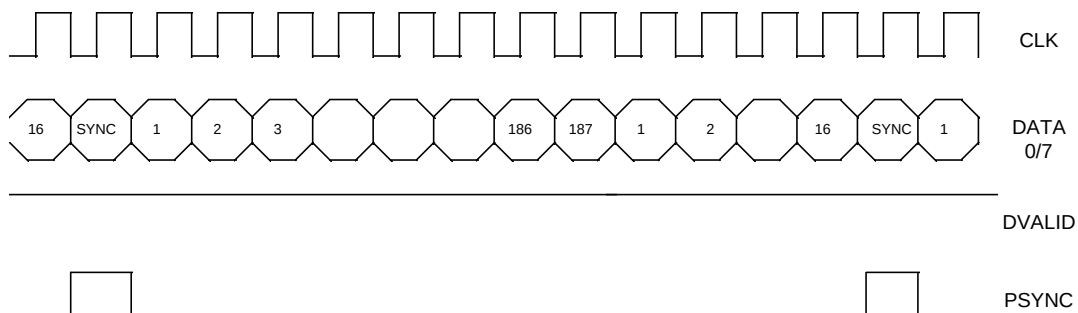
(*) packet timing : three transmission formats are supported, 188-byte packets, 204-byte packets without RS and 204-byte packets with RS (see 5.2.12.)



Transmission format with 188-byte packets



Transmission format with 204-byte packets without RS



Transmission format with 204-byte packets with RS

- for the RS422 interface :

This interface requires the interface module NTC/3320/ZG, NTC/3326/xx or NTC/3327/xE.

The use of a cable with individually shielded pairs is recommended. For operation up to 18 Mbit/s, the cable length should be limited to 2 m.

(*) signals and pin-out as used for interfacing with a Tiernan TD2 decoder

TIERNAN HD-15 PIN	RS422 CABLE		NTC 25 PIN
11	RT+	TTA	1
12	RT-	TTB	14
1	RD+	SDA	10
6	RD-	SDB	23
8	RR-	DTRA	11
3	RR+	DTRB	24
9, 15		GND	2, 15
		SHLD	13

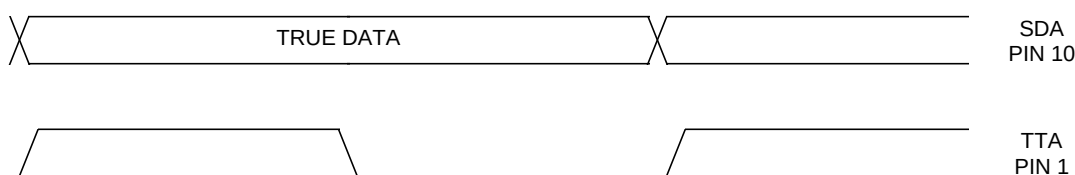
The cable can be ordered from NEWTEC, order code NTC/4147/AB.

(*) signals and pin-out as used for interfacing with a Tiernan TD3 decoder

TIERNAN HD-15 PIN	RS422 CABLE		NTC 25 PIN
12	RT+	TTA	1
11	RT-	TTB	14
1	RD+	SDA	10
6	RD-	SDB	23
8	RR-	DTRA	11
3	RR+	DTRB	24
9, 15		GND	2, 15
		SHLD	13

The cable can be ordered from NEWTEC, order code NTC/4147/AC.

(*) timing : normal (as shown below)
inversion is programmable in test mode.



Normal Clock-Data relationship

- for the HSSI interface : see datasheets of the corresponding module

This interface requires the interface module NTC/3362/xG.

4.8. Coaxial Interface

Via F BNC connector.

The type of cables and the connectors depend on the used interface module NTC/33xx/xx and are described in the data sheet of this module.

- for the DVB- ASI electrical interface :

This interface requires the interface module NTC/3327/xE.

The DVB-ASI electrical interface is conform to EN 50083-9 "Interfaces for CATV/SMATV Headends and Similar Professional Equipment".

(*) output connectors : BNC female

(*) signal flow : see data sheet

(*) electrical : 270 Mbaud NRZ
transformer coupled
75 Ω
sensitivity

(*) line code and synchronization : as per EN 50083-9

- for the DVB- ASI optical interface :

This interface requires the interface module NTC/3327/OE.

The DVB-ASI optical interface is conform to EN 50083-9 "Interfaces for CATV/SMATV Headends and Similar Professional Equipment".

- for the G.703 interface : see datasheets of the corresponding module

This interface requires the interface module NTC/3326/xx or NTC/2262/xG.

5. LOCAL MONITOR AND CONTROL

5.1. System Menu

The system menu provides information about the general device functions, the surrounding equipment and the used hardware. These parameters have to be set during installation of the modulator and saved in permanent memory. Hardware related parameters can only be changed in test mode.

When the modulator operates with a down-converter module (NTC/3631/LZ) all references to RF must be read as IF. This applies for this manual, the display indications and the remote messages.

5.1.1. Load Configuration

load configuration : system menu	select 0 - 4
-------------------------------------	--------------

The selection of one of the previously saved configuration from permanent memory. Such a configuration is formed by the control parameters as described in section 5.2..

Up to **five configurations** (0 up to 4) are available. At power-on or reset, the device automatically loads configuration 0.

In test mode five additional configurations (5 up to 9) become available, for test configurations only.

5.1.2. Save Configuration

save present config. as : system menu	select 0 - 4
--	--------------

Allows the operator to save the present control parameters (see 5.2.) as a configuration in permanent memory. Up to **five configurations** (0 to 4) are available. Save the default configuration in memory slot 0. It is not possible to save configurations that contain an incompatibility.

5.1.3. Test Functions

test functions : disabled
system menu

These are a number of additional functions, which are used for installation, test and problem solving. Some control parameters can be changed to a value not specified for normal operation.



WARNING :

If test functions are enabled, transmit suppression due to an active alarm is disabled.

Normal operation is not guaranteed when test functions are enabled. Damage can be done to this modulator and other equipment when the test functions are used by unqualified personnel or in an operational environment.

A password is required to enable test functions. This password is the NTC-number of the modulator (see 5.1.8.).

Procedure :

When this screen is displayed, type in the password "2063", the display will now show:

test functions : disabled
system menu toggle with <enter>

To enable press **E**

test functions : enabled
system menu toggle with <enter>

To disable press **E** again

5.1.4. Alarm Mask

alarm mask : 0000 hex
system menu

The alarm mask allows the operator to suppress the guarding of a dedicated alarm by setting the corresponding bit. The alarm mask must be entered during test mode.



-WARNING: the suppression of an alarm may prevent normal operation of the demodulator when the considered alarm occurs. Damage can be done to this demodulator and other equipment when the alarms are suppressed by unqualified personnel or in an operational environment.

Mask definition (bit 15 = MSB):

bit 15	: not used
bit 14	: Input level saturation
bit 13	: Buffer underflow
bit 12	: Buffer overflow
bit 11	: Device temperature
bit 10	: Data interface
bit 9	: Clock PLL
bit 8	: Synthesizer
bit 7	: not used
bit 6	: not used
bit 5	: not used
bit 4	: not used
bit 3	: -15.0 Volt Power Supply
bit 2	: +15.0 Volt Power Supply
bit 1	: -5.0 Volt Power Supply
bit 0	: +5.0 Volt Power Supply

It is strongly recommended to use 000000 hex as alarm mask, all alarms monitored.

Example : Suppose you want to mask the -15.0 Volt Power supply alarm.

Alarm mask = (0000 0000 0000 1000)_{BIN} = (0008)_{HEX}

5.1.5. Device Address for Serial Interface

device address for RMCP : 105
system menu new :

The device address for the NTC/2063/xx, used in the messages for remote serial M&C (see 6.), is a single byte with a value in the range 49 - 109. It identifies the device that has to handle the message from the remote control unit.



When the multi-user RS485 bus is used, each device on the bus must have a different address, unique in the system. If there are duplicate addresses on the same bus, one of the devices might block and a reboot would be necessary.

5.1.6. Device Control Mode

device control : local
system menu

The device control mode indicates via which interface the control parameters of the device may be changed. All status and monitoring information is always available at both interfaces.

In **local control** mode, all parameters can be changed via the front panel keypad and via RMCP (IEEE 488.1 style).

When the device operates in **remote control** mode, then all parameters may be changed via the remote interface (see 6.). But inputs from the front panel keypad are refused.

If the **local lock-out** state is selected, then the change of the device control mode is only allowed via RMCP.

If the device does not receive a request via RMCP within one minute, it changes its control state to local control, no lock-out.

5.1.7. Device Identification

Device : NTC/2063/IF High Rate DVB+8 Dem
Firmware : NTC/6094/AA v1.61 07 j01 2002

This is the device type (NTC-number) and sub-type (alphanumeric suffix) and identifies the hardware and firmware. The version number and the release date of the used firmware is displayed on the bottom line.

5.1.8. Device Serial Number

serial number : 01091540
system menu

The serial number of the device as known by the firmware. This number should be the same as the serial number on the back panel label.

5.1.9. Device Capability

device capability : FFFFFFFF
system menu

The device capability indicates the supported functionality and is needed for test and factory support.

5.1.10. Interface identification

interface : NTC/3389/AA ASI/SSI
system menu

This is the type (NTC-number) and sub-type (alphanumeric suffix) of the installed interface.

5.1.11. LNB Power Supply Control

LNB power supply : A disab. - B N.A.
system menu

The power supply voltage for the LNBs can be provided by the demodulator. It is available when **enabled** for the corresponding L-band input. Remark that the power is only switched on if the corresponding input is selected (see 5.2.6.).

When enabled, an LNB can be controlled via the level and an additional carrier for polarization and band selection according to the universal LNB for ASTRA satellites (see 5.2.4. & 5.2.5.).

The LNB power supply control is not applicable for the IF input (NTC/2063/lx only).

This parameter can only be changed in test mode.

5.1.12. BER Alarm

BER Alarm : disabled
system menu

In this menupart the BER alarm can be switched on and off. If the bit error ratio is above a certain predefined level an alarm will be generated, when in the "on" state. This will generate an alarm interface contact closure and RMCP alarm on the display (in alarm menu). The alarm can also be read via RS232/485. This parameter can only be changed in test mode.

5.1.13. BER Alarm Threshold

BER Alarm Threshold : 1000 E-6
system menu

The user can determine a BER threshold value between 10 and 1000 (units being 1E-6 errors per bit) or errors per million bits. This parameter can only be changed in test mode.

5.2. Control Menu

The control menu allows the operator to monitor the main device control parameters. If the modulator operates under local control, then they may also be changed via this menu. All the control parameters together form a configuration that can be stored in permanent memory (see 5.1.1. and 5.1.2.).

Some selections can be inaccessible to the operator. These selections become available in upgraded devices (see data sheets, contact Newtec Cy for upgrade options).

5.2.1. Interface rate

interf. rate (188) : 8.294118 Mbps control menu new: . Mbps

The interface rate of the device is coupled to the symbol rate (see 6.2.) in the following way :

$$\begin{aligned} \text{symbol rate} &= \text{interface rate} \\ &\quad * \text{framing overhead} \\ &\quad * 1/\text{RS-rate} \\ &\quad * 1/\text{FEC-rate} \\ &\quad * \text{modulation factor} \end{aligned}$$

$$\text{with : } - \text{framing overhead} = \frac{188}{187} \quad \text{for internal MPEG framing (see 5.2.12.)}$$

$$1 \quad \text{for external MPEG framing}$$

$$\begin{aligned} - \text{RS-rate} &= \frac{204}{188} \quad \text{for external MPEG framing with 188 byte frames and} \\ &\quad \text{internal MPEG framing (see 5.2.12.)} \end{aligned}$$

$$1 \quad \text{for external MPEG framing with 204 byte frames}$$

$$- \text{FEC-rate} = 1/2, 2/3, 3/4, 5/6, 6/7, 7/8 \text{ or } 8/9 \text{ (see 5.2.3.)}$$

$$\begin{aligned} - \text{modulation index} &= \frac{1}{2} \quad \text{for QPSK (see 5.2.3.)} \\ &\quad \frac{1}{3} \quad \text{for 8PSK} \\ &\quad \frac{1}{4} \quad \text{for 16QAM} \end{aligned}$$

Examples :

- (*) interface rate : 8.294118 Mbit/s, QPSK modulation, FEC-3/4, internal MPEG framing

$$\text{symbol rate : } 8.294118 \text{ Mbit/s} \times \frac{188}{187} \times \frac{204}{188} \times \frac{4}{3} \times \frac{1}{2} = 6.032086 \text{ Mbaud}$$

- (*) interface rate : 8.294118 Mbit/s, QPSK modulation, FEC-3/4, external MPEG framing, 188 byte frames

$$\text{symbol rate : } 8.294118 \text{ Mbit/s} \times 1 \times \frac{204}{188} \times \frac{4}{3} \times \frac{1}{2} = 6.000000 \text{ Mbaud}$$

- (*) interface rate : 8.294118 Mbit/s, QPSK modulation, FEC-3/4, external MPEG framing, 204 byte frames

$$\text{symbol rate : } 8.294118 \text{ Mbit/s} \times 1 \times 1 \times \frac{4}{3} \times \frac{1}{2} = 5.529412 \text{ Mbaud}$$

- (*) interface rate : 8.294118 Mbit/s, 8PSK modulation, FEC-5/6, external framing, 188 byte mode

$$\text{symbol rate : } 8.294118 \text{ Mbit/s} \times 1 \times \frac{204}{188} \times \frac{6}{5} \times \frac{1}{3} = 3.600000 \text{ Mbaud}$$

- (*) interface rate : 8.294118 Mbit/s, 16QAM modulation, FEC-3/4, external framing, 188 byte mode

$$\text{symbol rate : } 8.294118 \text{ Mbit/s} \times 1 \times \frac{204}{188} \times \frac{4}{3} \times \frac{1}{4} = 3.000000 \text{ Mbaud}$$

The interface rate is adjustable in steps of 1 bit/s, within the range of 1.290 Mbit/s up to 110.0 Mbit/s. The minimum and maximum value of the interface rate for a given configuration depends on the MPEG-framing mode (unframed, 188 byte or 204 byte frames as indicated), the FEC-rate and the modulation. These values are given in the table 5.2.1.

MODULATION / FEC RATE	INTERNAL MPEG		EXTERNAL MPEG 188 byte frames		EXTERNAL MPEG 204 byte frames	
	MIN	MAX	MIN	MAX	MIN	MAX
QPSK - FEC	1/2	1.290	41.250	1.296	41.470	1.407
	2/3	1.719	55.000	1.728	55.000	1.875
	3/4	1.934	55.000	1.944	55.000	2.110
	5/6	2.149	55.000	2.160	55.000	2.344
	6/7	2.210	55.000	2.222	55.000	2.411
	7/8	2.256	55.000	2.268	55.000	2.461
8PSK - FEC	2/3	2.579	82.500	2.592	82.941	2.813
	5/6	3.223	103.125	3.240	103.676	3.516
	8/9	3.438	110.000	3.456	110.000	3.750
16QAM - FEC	3/4	3.868	110.000	3.888	110.000	4.219
	7/8	4.512	110.000	4.536	110.000	4.922

Table 5.2.1. : Interface rate range (Mbit/s, rounded)

These limits are derived from the absolute maximum of the interface rate, 55.0 Mbit/s for QPSK and 110.0 Mbit/s for 8PSK-modulation and 16QAM, and the limitation on the symbol rate at 1.40625 - 45.0 Mbaud.

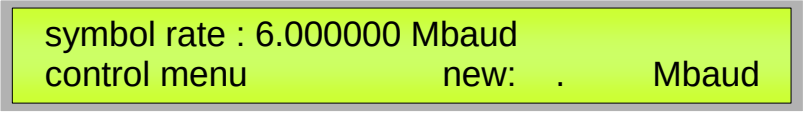
Depending on the ordered option (see data sheet), additional limitations can be imposed.

Remark also that the selection of some interfaces give additional restrictions for the interface rate (see data sheets).

The selection of an interface rate outside the allowed limits is indicated via an incompatibility.

The fixed coupling between the interface rate and the symbol rate requires that for optimal operation the satellite clock is used as interface clock. This selection keeps the contents of the deframer buffer at its center. Remark that for the other clock selections, external or internal clock, the buffer content changes proportional to the difference between the selected clock and the satellite clock. This can cause the loss of data (buffer overflow) or the insertion of dummy data (buffer underflow).

5.2.2. Symbol rate



symbol rate : 6.000000 Mbaud
control menu new: . Mbaud

The symbol rate is adjustable in steps of 1 baud, within the range of **1.40625 Mbaud up to 45.0 Mbaud**. Since the symbol rate is coupled to the interface rate, see 5.2.1. for examples, couplings and remarks.

5.2.3. FEC-rate and Modulation

QPSK modul. – FEC-rate 3/4 * 188/204
control menu change with + or -

The selection of the FEC-rate and modulation are coupled and depends on the present modulation standard of the device (see 5.2.17.). Possible selections:

MODULATION	FEC RATE	STANDARD
QPSK	1/2 * (188/204)	DVB and NTC
	2/3 * (188/204)	
	3/4 * (188/204)	
	5/6 * (188/204)	
	6/7 * (188/204)	
	7/8 * (188/204)	
8PSK	2/3 * (188/204)	DVB and NTC
	5/6 * (188/204)	
	8/9 * (188/204)	
16QAM	3/4 * (188/204)	DVB and NTC NTC only
	7/8 * (188/204)	

Table 5.2.3. Modulation & coding rate options

The selection of an FEC-rate and or modulation type, for which the present symbol rate is outside the allowed limits, is indicated via an incompatibility.

5.2.4. LNB A Control

LNB A control
control menu change with + or -

The LNB connected at L-band input A can be controlled via its power supply, according to the universal LNB for ASTRA satellites :

- **vertical** or **horizontal** polarization via the level, respectively 13 volt or 18 volt
- band selection, **low** or **high**, via an additional 22 kHz signal

The LNB A control is not applicable if the power supply for input A is disabled (see 5.1.11.).

5.2.5. LNB B Control

LNB B control
control menu

change with + or -

The LNB connected at L-band input B can be controlled via its power supply, according to the universal LNB for ASTRA satellites :

- **vertical** or **horizontal** polarization via the level, respectively 13 volt or 18 volt
- band selection, **low** or **high**, via an additional 22 kHz signal

The LNB B control is not applicable if the power supply for input B is disabled (see 5.1.11.) and for NTC/2063/Ix demodulators.

5.2.6. LNB Input

LNB select : input IF
control menu

change with + or -

The LNB input selection controls which input signal is passed to the demodulator. For a NTC/2063/Fx demodulator there are two L-band inputs, **A** and **B**, available, while a NTC/2063/Ix demodulator selects between an **L-band** input (A) and an **IF** input.

The NTC/2063/Ix demodulator indicates an incompatibility if the IF input is selected, while the IF(L) frequency is outside the IF range (see 5.2.8.).

5.2.7. Excess Bandwidth

excess bandwidth : 35% auto
control menu

toggle with <enter>

The excess bandwidth defines the RCRO factor for the matched filter at the demodulator input. The standard **35% RCRO** or the additional **25%** can be selected.

If the **auto** select mode is used for the choice of the excess bandwidth, then 35 % RCRO is chosen for the QPSK modulation modes and 25% RCRO for the 8PSK and 16 QAM modulation modes.

5.2.8. IF(L) Frequency

receive frequency : 70.00000 MHz
control menu new: MHz

The carrier frequency is adjustable in steps of 50 Hz, within the range of **920 MHz up to 2150 MHz** for a NTC/2063/Lx demodulator or **52.0 MHz up to 176.0 MHz** for a NTC/2063/lx demodulator. The NTC/2063/lx demodulator indicates an incompatibility if the IF input is selected, while the IF(L) frequency is outside the IF range (see 5.2.6.).

5.2.9. Carrier acquisition range

acquisition range : 500 kHz-ptp
control menu new: kHz

The carrier acquisition range (peak-to-peak) is adjustable in steps of 1 kHz, within the range **1.0 kHz to 5000.0 kHz**. The demodulator synchronization time is proportional to this acquisition sweep width.

5.2.10. Spectrum Inversion

spectrum inversion : OFF auto
control menu toggle with <enter>

The selection of the **auto** mode allows the demodulator to solve the spectrum ambiguity when the status of the received spectrum is unknown. When the status of the received spectrum is known, it is recommended to use this direct setting since it minimizes the acquisition time. The spectrum inversion **off** mode, or direct spectrum, corresponds with the INTELSAT specification IESS-308 (Rev.8 - pg. 18 & 69) and with the DVB standard ETS 300 421 (December 1994).

5.2.11. Descrambler

descrambler : ON
control menu

The descrambler is fixed **on** in operational mode, according to the DVB specifications. In test mode it can be switched **off**.

5.2.12. MPEG Framing

MPEG framing : extern, 188 bytes
control menu toggle with <enter>

The interface MPEG framing defines the data format sent at the interface.

DVB compliant transmission always requires **external** framing. This external framing can be **188 byte packets** or **204 byte packets**. The 204 byte packets contain beneath the SYNC-byte and the 187 data bytes also 16 dummy bytes, marked with data valid LOW, at the place of the RS parity bytes.

In combination with a NTC/2080/xx modulator, a non-DVB compliant transmission of continuous data can be realized by selecting the **internal** framing. Then the modulator inserts after each 187 data bytes a SYNC-byte, to generate a "pseudo" MPEG data stream. Take care that the modulator operates also with external transmit clock, to prevent the insertion of dummy MPEG frames.

The selection of a framing mode which is not supported by the selected interface is indicated via an incompatibility.

5.2.13. Deframer Buffer Size

average buffer size : 64 MPEG frames
control menu new : MPEG fram.

The average size of the deframer output buffer is adjustable in steps of 1 frame, within the range of **3 up to 128 MPEG frames**. This buffer must be taken large enough to compensate for Doppler variations and clock drift, when the external buffer output clock is selected. But remark that the delay of the data in this buffer is proportional to its size.

5.2.14. Deframer Buffer Output Clock

defr. buffer outp. clock : satellite
control menu toggle with <enter>

The deframer buffer output clock is also the clock for the data output at the interface.

The selection of the **satellite** clock slaves the output clock to the satellite received clock, as regenerated by the demodulator. If the demodulator is not synchronized then the device falls automatically back to the internal clock.

If a stable **external** clock at user rate is available, then it can be selected as buffer output clock. Remark that the external clock should be disconnected if not selected, in order to avoid interference with the selected clock on the output data stream. Some interfaces and/or interface boards does not support the use of an external clock (see data sheets). The selection of the external clock in such a case is flagged as an incompatibility.

The **internal** clock is a fixed fallback clock, which can also be selected for data output. The device automatically falls back to this clock if the satellite or the external clock is not present or not valid.

Take care that the buffer is large enough to compensate for Doppler variations and clock drift when the external or internal clock are selected.

5.2.15. Interface for Sub-D Connector

sub-D int. : DVB-SPI-LVDS
control menu toggle with <enter>

The selectable interfaces for the sub-D connector depends on the **type of interface module NTC/33xx/xx** in the device (see 5.1.10.).

Remark that some interface modules allow the selection of a sub-D and a coax interface at the same time.

The sub-D interface can be switched **off** if not needed. Remark that its is automatically switched off by the selection of some coax interfaces on certain interface modules or if the used module does not have a sub-D interface (see data sheets).

Some interfaces can only be used over a limited interface rate (see data sheet). The selection of an invalid interface and interface rate combination is flagged as an incompatibility. An incompatibility is also indicated if the selected interface is not available on the present interface module or if the external clock is selected for buffer output and this clock is not supported by this interface.

5.2.16. Interface for coaxial connector

coax 1-2 : electr. ASI
control menu toggle with <enter>

coax 3 : SSI
control menu toggle with <enter>

The selectable interfaces for the coax connector depends on the **type of interface module NTC/33xx/xx** in the device (see 5.1.10.).

Some interfaces can only be used over a limited interface rate (see data sheet). The selection of an invalid interface and interface rate combination is flagged as an incompatibility. An incompatibility is also indicated if the selected interface is not available on the present interface module or if the external clock is selected for buffer output and this clock is not supported by this interface.

5.2.17. Modulation Standard

modulation standard : DVB
control menu toggle with <enter>

The modulation standard defines the major operating mode of the device.

The **DVB standard EN 301210** uses the following settings :

- QPSK, 8PSK or 16QAM modulation
- FEC-encoding enabled
- DVB scrambling
- interleaver enabled

The **NTC standard, EN 300421** for QPSK modulation and **NTC-proprietary** for 8PSK and 16QAM modulations, uses the following settings :

- QPSK, 8PSK or 16QAM modulation
- FEC-encoding enabled
- DVB scrambling
- interleaver enabled

Switching between the different modulation standards changes the corresponding control parameters as required.

16QAM modulation in combination with FEC-rate 7/8 is not supported for the DVB standard. The selection of this combination is indicated as an incompatibility.

5.3. Alarm Menu

The modulator has 2 indications for each guarded function: status and event counter.

The status indicates the actual situation of the guarded function:

- ◆ O.K. : the function operates as expected
- ◆ ALARM : a malfunction is detected
- ◆ memorized alarm : a malfunction has been detected, but has now disappeared
- ◆ not applicable : the guarding of this alarm is not needed for the present settings
- ◆ masked : the guarding of this alarm is suppressed by the operator

The event counter indicates the number of times that an alarm has occurred since power-on or the last counter reset. The occurrence of an alarm is the status transition from "O.K." to "ALARM". The counter saturates at 255 events.

5.3.1. Reset All Memorised Alarms

reset all memorised alarms with <clear>
alarm menu reset counter with <clear>

Pressing the <C>-key (clear) resets all memorized alarms and event counters.

5.3.2. Incompatibility Indication

control setting : OK
alarm menu reset counter with <clear>

The device gives an incompatibility indication if a new control setting is in conflict with one or more of the other present settings.

The type of incompatibility is indicated by a number, which is to be interpreted as a bitmap (bit 7 = MSB). The corresponding bit is set if there is an incompatibility between :

- bit 7 : control setting and device capability
- bit 6 : modulation standard and FEC/modulation mode
- bit 5 : selected input and IF(L) frequency
- bit 4 : selected interface and MPEG framing
- bit 3 : selected interface and clock selection
- bit 2 : selected interface and interface module
- bit 1 : selected interface and interface rate
- bit 0 : symbol rate and FEC/modulation mode

5.3.3. +5.0 Volt Power Supply

+4.97 V pow.sup. no alarm	number 0
alarm menu	reset counter with <clear>

A +5.0 volt power supply alarm indicates that this power supply exceeds the +4.80 volt or +5.20 volt limit. The hardware power supply circuit generates a hardware reset if this power supply drops below +4.60 V or rises above +5.40 V.

5.3.4. -5.0 Volt Power Supply

-5.09 V pow.sup. no alarm	number 0
alarm menu	reset counter with <clear>

A -5.0 volt power supply alarm indicates that this power supply exceeds the -5.50 volt or -4.50 volt limit.

5.3.5. +15.0 Volt Power Supply

+15.07 V pow.sup. no alarm	number 0
alarm menu	reset counter with <clear>

A +15.0 volt power supply alarm indicates that this power supply exceeds the +14.0 volt or +16.0 volt limit.

5.3.6. -15.0 Volt Power Supply

-14.98 V pow.sup. no alarm	number 0
alarm menu	reset counter with <clear>

A -15.0 volt power supply alarm indicates that this power supply exceeds the -16.0 volt or -14.0 volt limit.

5.3.7. Device Temperature

dev. temperature	no alarm	number 0
alarm menu	reset counter with <clear>	

A device temperature alarm indicates that this temperature exceeds the +10° C or +70° C limit.

5.3.8. Synthesiser

synthesiser	no alarm	number 0
alarm menu	reset counter with <clear>	

Indicates a failure in the hardware that generates the L-band carrier frequency.

5.3.9. Clock PLL

clock PLL	no alarm	number 0
alarm menu	reset counter with <clear>	

The clock PLL alarm is generated when the symbol clock PLL is not locked to its reference.

5.3.10. Input Level Saturation

level saturation	no alarm	number 0
alarm menu	reset counter with <clear>	

The input level saturation alarm indicates that the received level is too high. This results in a degradation of the performance and even the impossibility to synchronize. If the alarm occurs, reduce the transmit level or insert attenuators before the demodulator.

5.3.11. Buffer Overflow

buffer overflow	no alarm	number 0
alarm menu	reset counter with <clear>	

This alarm indicates that the deframer buffer contains too much data. An overflow causes the loss of a whole number of MPEG frames, so the buffer is re-centered.

5.3.12. Buffer Underflow

buffer overflow	no alarm	number 0
alarm menu	reset counter with <clear>	

This alarm indicates that the deframer buffer contains too less data. An overflow causes the insertion of a whole number of dummy MPEG frames, so the buffer is re-centered.

5.3.13. Data Interface

data interface	no alarm	number 0
alarm menu	reset counter with <clear>	

The data interface alarm indicates a malfunction of the interface module or that no valid input clock is received via this module. The verified functions depend on the selected outputs and contain among others : clock synchronization, coding status, interface buffer over/underflow and power presence (see data sheets), just as the presence of the external clock, if selected.

5.3.14. Demodulator Sync

demodulator sync	no alarm	number 0
alarm menu	reset counter with <clear>	

An "alarm" is indicated if the demodulator is not synchronized on the incoming carrier.

5.3.15. Viterbi Decoder Sync

Viterbi dec. sync	no alarm	number 0
alarm menu	reset counter with <clear>	

An "alarm" is indicated if the Viterbi decoder is not synchronized on the incoming data stream. This indication is not applicable as long as the demodulator is not synchronized or when QPSK modulation is selected.

5.3.16. Unique Word Detection

unique word det.	no alarm	number 0
alarm menu	reset counter with <clear>	

An “alarm” is indicated if the unique word is not detected in the incoming data stream. This indication is not applicable as long as the Viterbi decoder is not synchronized.

5.3.17. Reed-Solomon Decoder Sync

RS-decoder sync	no alarm	number 0
alarm menu	reset counter with <clear>	

An “alarm” is indicated if the RS decoder is not synchronized on the incoming data stream. This indication is not applicable as long as the Viterbi decoder is not synchronized.

5.3.18. Deframer Sync

deframer sync	no alarm	number 0
alarm menu	reset counter with <clear>	

An “alarm” is indicated if the MPEG framing is not detected in the incoming data stream. This alarm coincides with the unique word detection for the present implementation. This indication is not applicable as long as the Viterbi decoder is not synchronized.

5.3.19. BER Threshold

BER threshold	no alarm	number 0
alarm menu	reset counter with <clear>	

An “alarm” is indicated if the bit error ratio of the received signal is larger than the BER alarm threshold preset value (See also 5.1.12 and 5.1.13). This gives the possibility to have a “pre-warning alarm” whenever the BER becomes dangerously close to an unlock situation. Based on the contact closure or the RMCP message, preventive actions can then be taken.

5.4. Monitor Menu

The monitor menu gives information about variables measured in the device. These variables can not be modified by the operator.

5.4.1. Internal temperature

internal temperature : +37.8 ° C
monitor menu

The internal temperature of the device must be within +10 and +70° C (see 5.3.7.). Typical, this temperature should be around 40°C (± 10°C).

5.4.2. Receive Level

receive level : -45.20 dBm (-115.20 dBm/Hz)
monitor menu

The received level is given in dBm. Also the Co-level density is indicated in dBm/Hz

The maximum wanted input level specified is :

$$Co]_{MAX.} + 10\log (f_{baud})_{MAX.} = -109 \text{ dBm/Hz} + 10 \log (45.0 \text{ Mbaud}) = -32.5 \text{ dBm.}$$

The maximum total input power at the Demodulator input in band (920 - 2150 MHz) is specified as -17 dBm.

Saturation is indicated with < (lower) or > (upper). Upper level saturation generates the input level saturation alarm. (see 5.4.10.).

5.4.3. Eb/No Estimation

Eb/No-estimation : 15.20 dB
monitor menu

The measured Eb/No estimation in dB is used for link margin estimation (see 5.4.9.) and can be used for link degradation estimation.

5.4.4. Carrier Frequency Offset

carrier frequency offset : 1.200 kHz
monitor menu

The measured carrier frequency offset in Hz is given against the selected IF(L) frequency (see 5.2.8.).

5.4.5. Symbol Rate Offset

symbol rate offset : -05.8 Hz
monitor menu

The measured symbol rate offset in Hz is given against the selected rate (see 5.2.2.).

5.4.6. Symbol Error Rate before FEC

symb. error ratio before FEC: 1.20 E-6
monitor menu

The symbol error rate of the received signal as it is measured by the Trellis decoder. The symbol error rate indication is not applicable for QPSK-modulation.

5.4.7. Byte Error Rate before RS-decoder

byte error ratio before RS : 1.3 E-9
monitor menu

The byte error rate as measured at the input of the Reed-Solomon decoder. The byte error rate indication is not applicable if the RS-decoder is not synchronized. Remark that for some FEC-rates with QPSK modulation, there can be a residual error rate of 1 to 5E-6. These errors does not influence the general performance of the device.

5.4.8. Bit Error Rate after decoding

bit error ratio after RS : 12.5 E-12
monitor menu

The estimated bit error rate at the output data as calculated from the pre-RS decoder byte error rate.

5.4.9. Link Margin

Eb/No margin for BER 1.0E-9 : 6.24 dB
monitor menu

The estimated amount of Eb/No in dB, that the link may degrade before BER > 1E-9.

This margin is calculated from the indicated Eb/No-estimate and the typical user Eb/No (see data sheet).

5.4.10. PRBS Bit Error Rate

PRBS BER : N.A. errors/sec
monitor menu

This is the measured bit error rate of the received PRBS signal. This figure is not applicable if the PRBS is not received (not detected in the incoming signal). Remark that "all-ones" AIS are also treated as a PRBS signal. This is indicated by "ones" on the LCD.

5.4.11. PRBS Bit Error Count

PRBS : not received
monitor menu

The number of errors are counted on the received PRBS signal since detection of the PRBS signal or since last counter reset by the operator. The given indications are :

- the number of errors accumulated (differences)
- the number of seconds that the errors are counted

These figures are not applicable if the PRBS is not received (not detected in the incoming signal).

Remark that "all-ones" are also treated as a PRBS signal. This is indicated by "ones" on the LCD and in the engineering string.

5.4.12. Buffer Contents

buffer contents : 23 MPEG frames
monitor menu

The content of the data output buffer given in MPEG frames.

Buffer underflow or overflow (see 5.3.11 & 5.3.12.) are indicated when the buffer contains less than one frame or respectively is filled up to less than one frame from its maximum size (double average). Then the buffer is centered.

When the satellite clock is selected as buffer output clock, then a loop maintains the buffer content at the average buffer size (see 5.2.13.).

5.4.13. Internal Test Variables (Test Mode Only)

The following variables can be monitored in test mode only :

- I-channel DC-offset
- Q-channel DC-offset
- Level Detector
- Unfiltered Eb/No Estimator
- Carrier Correlator
- Clock Correlator I Component
- Clock Correlator Q Component
- Clock Correlator
- Unfiltered Sync Detector
- Filtered Sync Detector
- Synthesizer Lock Voltage
- Sample clock PLL offset
- Buffer Output Clock Phase Error
- Buffer Output Clock Offset

Please contact NEWTEC Cy for information and interpretation of these variables when needed.

5.5. Test Menu

The test menu contains a number of assistance functions for test and maintenance of the device.

This menu is only available when the test functions are enabled. If the test functions are disabled, then the values of these parameters are set to the default values for normal operation.

Remark that it is not guaranteed that the device operates according to specifications when these parameters are changed. So these functions may never be used in an operational environment nor changed by unauthorized personnel.

- Level Control
- Carrier Frequency Control
- Clock Frequency Control
- Side Band Detection
- Test D/A-Converter Selection
- 480.0 MHz LO Frequency
- Tuner Control
- Carrier Correlator
- Clock Correlator
- Data Clock Edge
- HSSI Interface Control
- Interface Status and Alarms
- Initialize LCA's
- Serial Interface
- Interface Identification
- Interface Table Read

Please contact NEWTEC Cy for information and use of these functions when needed.

6. REMOTE MONITORING AND CONTROL

A master control unit, typically a Personal Computer, can supervise one or multiple devices via the Remote Monitoring and Control Protocol (RMCP), a message oriented protocol.

6.1. Interface Hardware

The device contains the hardware for the RS485 and RS232 interface. The operator must indicate which hardware to use via the straps near the serial M&C connector on the main board (see 7.5.).

WARNING ! The use of the wrong hardware can damage this device, the remote control unit and possible other devices on the bus.

6.1.1. RS485 Hardware

The RS485 interface is a multi-user serial interface, which allows one master control unit to supervise several devices in parallel.

The pinout on the 9-pin sub-D connector for RS485 is:

Pin 1	shield (ground)
Pin 2	- (not connected)
Pin 3	send data A (from control unit to device)
Pin 4	receive data A (by control unit, send from device)
Pin 5	signal ground
Pin 6	receive data B (by control unit, send from device)
Pin 7	- (not connected)
Pin 8	- (not connected)
Pin 9	send data B (from control unit to device)

The polarities of receive data and send data:

Data logic 1	: $V_a < V_b$
Data logic 0	: $V_a > V_b$

The control signals are not used for RS485. Typical electrical levels for V_a and V_b are 0 volt and 5 volt.

6.1.2. RS232 Hardware

The RS232 interface is a point-to-point serial interface, which allows one master control unit to supervise only one device.

The pinout on the 9-pin sub-D connector for RS232 is:

Pin 1	shield (ground)
Pin 2	receive data (RD, by device, send from control unit)
Pin 3	transmit data (TD, from device to control unit)
Pin 4	data terminal ready (DTR)
Pin 5	signal ground
Pin 6	- (not connected)
Pin 7	request to send (RTS)
Pin 8	clear to send (CTS)
Pin 9	- (not connected)

The polarity of the signals:

Data logic 1 : $V_{data} < 0\text{ V}$
Data logic 0 : $V_{data} > 0\text{ V}$

Control logic 1 : $V_{sig} > 0\text{ V}$
Control logic 0 : $V_{sig} < 0\text{ V}$

Typical electrical levels are between 3 volt and 12 volt.

6.1.3. Serial Interface - Line Settings

The main line settings for this serial interface are :

Asynchronous data transfer
1 start bit (logic "0")
7 data bits (LSB first on line)
Even parity
1 stop bit (logic "1")
9600 baud

Remark that there is no flow control on the serial interface. Next to correct formatted messages, the only significant character is the SYNC-character (value 16 hex), which is sent by the device to indicate that it is busy with executing the command and preparing the response. This prevents other devices from taking the bus if the response can not be given immediately.

6.2. Message Exchange Protocol

The control unit sends a “request” message to a device, identified by its unique address. The addressed device interprets the message, performs the requested action and sends back a “response” message.

The receiving device rejects all messages with transmission errors without any further action. Transmission errors are: no stop bit, parity error, LRC-error and message receive buffer overflow.

All correct formatted messages, except some special system messages, are responded by the addressed device with an acknowledge message. Only in a few restricted cases, the device does not respond to a request from the control unit. This is for example the case when a general device reset is requested.

Correct received messages, which can not be handled by the device are refused via a no-acknowledge “error” message, containing the reason why the message is rejected.

Remark that a device never sends messages on its own initiative. It only responds to requests from the control unit. Thus to maintain a complete overview of the system, it has to question each device regularly.

The total transmit time of a complete message may not exceed 250 ms. If the message is not completed within this time, then the remainder of it is supposed being lost and the message is dropped without any further action.

6.3. Message Format

The general syntax for all messages is:

Start byte
Address byte
Message header
Message data
End of text byte
Checksum byte

6.3.1. Start Byte

The start byte defines unequivocal the beginning of a new message.

There are three possible start bytes:

- 02 hex: start-of-text (STX), used for a request send by the control unit to a device
- 06 hex: acknowledge (ACK), used for the response from a device to the control unit
- 15 hex: no-acknowledge (NAK), used for the "error" response from a device to the control unit.

6.3.2. Address Byte

The address identifies the destination device for the request message and the source device for the response message. Only the device that recognizes its own address may treat the message. Remark that each device must have a unique address when the control unit guards more than one device via a multi-user bus.

Valid addresses are in the range 49 (31 hex, ASCII "1") up to 110 (6E hex, ASCII "n").

For a limited set of messages, also the RMCP-broadcast address 111 (6F hex, ASCII "o") may be used. This address is reserved for (test) systems with only one device connected to the control unit. Messages that are handled at the broadcast address are the device identification and the RMCP-address.

Multiple identical parts, which use the same messages, in a device are distinguished via sub-addressing. A colon precedes a sub-address ":" (ASCII code 3A hex).

Valid sub-addresses are in the range 49 (31 hex, ASCII "1") up to 110 (6E hex, ASCII "n").

Thus with sub-addressing, the complete message format becomes:

`<stx><addr>:<sub-addr><header><data><etx><lrc>`

The "broadcast" sub-address "o" (ASCII code 6F hex) indicates that the command applies to all parts in the device, which use the corresponding message. This can be used for commands only. When a monitoring value is requested via the broadcast sub-address, then the value for the first part in the sub-address range is given.

6.3.3. Message Header

The header defines the contents of the message and thus how the device should handle it.

Most headers exist of three capital or minor ASCII characters (respectively 41 hex up to 5A hex and 61 hex up to 7A hex). Remark that the header syntax is case sensitive.

For some special purpose messages, a one-character header is used. This is an ASCII representable character in the range 20 hex up to 7F hex, capital and minor letters excluded.

6.3.4. Message Data

The message text is a string of ASCII representable characters (20 hex to 7F hex). The meaning and the size of this string are defined by the header.

Possible message text is :

- no data, this is indicated as the "null"-string
- a query for information, indicated by a question mark "?". Other possible queries are minimum value ">", maximum value "<" and with reset "!"
- an enumeration, selected from a limited set of alternatives
- a numerical value, with the format and units as given for each command. Remark that format and size are fixed. Example "s##.#" for sign, two digits, decimal dot and one digit
- a text string, with fixed or variable length
- a dummy argument, this is a string of question marks, which replaces the normal value when the requested parameter is unknown
- a not applicable indication, this is a string of "#"-signs, which replaces the normal value when the requested parameter is not applicable.

6.3.5. End of Text Byte

The end of text byte indicates the end of the message text. This is the "end-of-text" character, with as value 03 hex.

This is also the last but one byte of the complete message. It is only followed by the checksum.

6.3.6. Checksum Byte

The last byte of each message is a checksum. This is a one-byte longitudinal redundancy check (LRC). It is calculated as the bitwise exclusive-OR of all the message bytes, including the message delimiters (STX, ETX, ...) and the address.

The bitwise exclusive-OR function can be calculated by writing the binary value of all the concerned bytes beneath each other and then performing a modulo-2 addition on each column, thus without carry propagation (see examples).

6.3.7. Examples

For a device with RMCP address 51 (33 hex, ASCII "3")

- request device identification : "0?"

<STX>30?<ETX>=

<STX>	02 hex	000 0010	
3	33 hex	011 0011	
0	30 hex	011 0000	
?	3F hex	011 1111	
<ETX>	03 hex	000 0011	
<LRC>	3D hex	011 1101	thus ASCII "="

- select 8PSK modulation and FEC-rate 5/6 : "RMx85"

<STX>3RMx85<ETX>X

<STX>	02 hex	000 0010	
3	33 hex	011 0011	
R	52 hex	101 0010	
M	4D hex	100 1101	
x	78 hex	111 1000	
8	38 hex	011 1000	
5	35 hex	011 0101	
<ETX>	03 hex	000 0011	
<LRC>	58 hex	101 1000	thus ASCII "X"

- select electrical ASI interface : ":2RIs3", thus the coaxial interface at sub-address 2

<STX>3:2RIs3<ETX>a

<STX>	02 hex	000 0010	
3	33 hex	011 0011	
:	3A hex	011 1010	
2	32 hex	011 0010	
R	52 hex	101 0010	
I	49 hex	100 1001	
s	73 hex	111 0011	
3	33 hex	011 0011	
<ETX>	03 hex	000 0011	
<LRC>	61 hex	110 0001	thus ASCII "a"

6.4. Remote M&C Commands

The header is represented in “**bold**” and the data in “*italic*” for each possible message. The start byte (STX, ACK or NAK), the address byte, the end of text byte (ETX) and the checksum byte are not given. Their use follows the standard rules unless explicitly given, i.e. the sub-address if needed.

Remark that the values send in a response are always the parameters as used by the device. They can differ from the requested values because of internal rounding, saturation or other limitations.

The present alarm status indicates if an alarm has occurred since the previous request. Thus it is required that the control unit regularly interrogates each device, to obtain an accurate overview of their status.

The “!”-query of a memorized alarm or state resets the memorized value.

6.4.1. System Parameters

6.4.1.1. Load Configuration

<enum> :	0	load done
	1	load failed (not loaded)
	2	load errors (LRC-error)
	3	configuration # does not exist

request :	SCg#	load configuration #
response :	SCg <enum>	load result

6.4.1.2. Save Configuration

<enum> :	0	save done
	1	save refused (configuration error or incompatibility)

request :	SCs#	save present settings as configuration #
response :	SCs <enum>	save result

6.4.1.3. Test Functions

<enum> :	0	disabled (operational mode)
	1	enabled

request :	SMm?	get present status
	SMm <enum>	change status
response :	SMm <enum>	present or new status

request : **SLs?** get serial number of the device
 response : **SLsyymddnn** serial number of the device

6.4.1.9. Device Capability

request : **SLc?** get device capability
 response : **SLchhhhhh** device capability

6.4.1.10. Interface Identification

request : **Isi?** get interface identification
 response : **Isi## 3xxxxx #####** interface identification

6.4.1.11. LNB Power Supply Control

<enum> : 0 A and B disabled
 1 A enabled - B disabled
 2 B enabled - A disabled
 3 A and B enabled

request : **XVs?** get present LNB power supply status
 XVs<enum> change LNB power supply status as requested

response : **XVs<enum>** send present or new LNB power supply status

6.4.1.12. Self Test

request : **SSs?** perform self test and send result
 Sss? get self test result
 SsS? get self test result
 SsS! get self test result with reset

response : **SSs###** self test result
 Sss### self test result
 SsS### self test result

Remark that a self test disturbs normal operation. To prevent inconvenience, transmit is switched off during this test. Afterwards, normal operation is resumed.

6.4.1.13. Device Reset

request : **SRr!** device reset

response : no response !!

After the reset, the "had reset" flag is activated.

Remark that during the device reset no messages are handled.

6.4.1.14. Device Reset Flag

<enum> : 0 no device reset since last flag clear
 1 device power-on or reset (since last flag clear)

request : **Srs?** get device reset flag
 SrS? get device reset flag
 SrS! get device reset flag with flag clear

response : **Srs<enum>** device reset flag
 SrS<enum> device reset flag

6.4.2. Control Parameters

6.4.2.1. Interface Rate

request : **RRr?** get present rate
 RRr##### change rate

response : **RRr#####** present or new rate in bit/s

6.4.2.2. Symbol Rate

request : **RRs?** get present rate
 RRs##### change rate

response : **RRs#####** present or new rate in baud

6.4.2.3. FEC-rate and Modulation

<enum1> : 1 QPSK modulation
 8 8PSK modulation
 6 16 QAM modulation

<enum2> : 0 no FEC (SKYPLEX only)
 1 rate 1/2 * 188/204
 2 rate 2/3 * 188/204
 3 rate 3/4 * 188/204
 5 rate 5/6 * 188/204
 6 rate 6/7 * 188/204
 7 rate 7/8 * 188/204
 8 rate 8/9 * 188/204

request : **RMx?** get present modulation type and FEC rate
 RMx<enum1><enum2> change modulation type and FEC rate
 RMm? get present modulation type
 RMf? get present FEC rate

response : **RMx<enum1><enum2>** present or new modulation type and FEC rate
 RMm<enum1> present modulation type
 RMf<enum2> present FEC rate

6.4.2.4. LNB A Control

<enum> : 0 vertical - low band
 1 vertical - high band
 2 horizontal - low band
 3 horizontal - high band

request : :1XBm? get present status
 :1XBm<enum> change status

response : :1XBm<enum> present or new status

6.4.2.5. LNB B Control

<enum> : 0 vertical - low band
 1 vertical - high band
 2 horizontal - low band
 3 horizontal - high band

request : :2XBm? get present status
 :2XBm<enum> change status

response : :2XBm<enum> present or new status

6.4.2.6. LNB Input

<enum> : 1 input A
 2 input B

request : XBi? get present status
 XBi<enum> change status

response : XBi<enum> present or new status

6.4.2.7. Excess Bandwidth

<enum> : 0 auto (25% for QPSK and 35% for 8PSK - 16QAM)
 1 35% RCRO
 2 25% RCRO

request : RFr? get present status
 RFr<enum> change status

response : RFr<enum> present or new status

6.4.2.8. IF(L) Frequency

request :	RFf?	get present frequency
	RFf>	get minimum frequency
	RFf<	get maximum frequency
	RFf#####	change frequency
response :	RFf#####	present or new frequency, or limit in Hz

6.4.2.9. Carrier Acquisition Range

request :	RDr?	get present range
	RDr>	get minimum range
	RDr<	get maximum range
	RDr#####	change range
response :	RDr#####	present or new range, or limit in Hz

6.4.2.10. Spectrum Inversion

<enum> :	0	auto select
	1	direct spectrum
	2	inverted spectrum
request :	RMi?	get present status
	RMi<enum>	change status
response :	RMi<enum>	present or new status

6.4.2.11. Descrambler

<enum> :	0	disabled (test only)
	1	enabled
request :	RMs?	get present status
	RMs<enum>	change status
response :	RMs<enum>	present or new

6.4.2.12. MPEG Framing

<enum> :	0	external MPEG framing, 188 byte mode
	1	internal MPEG framing
	2	external MPEG framing, 204 byte frames, without RS
	3	external MPEG framing, 204 byte frames, with RS (not applicable)
request :	RMe?	get present status
	RMe<enum>	change status
response :	RMe<enum>	present or new status

6.4.2.13. Deframer Buffer Size

request :	RIb?	get present size in bytes
	RIb>	get minimum size in bytes
	RIb<	get maximum size in bytes
	RIb#####	change size in bytes
	RIf?	get present size in frames
	RIf>	get minimum size in frames
	RIf<	get maximum size in frames
	RIf###	change size in frames
	RIb#####	present or new size, or limit in bytes
	RIf###	present or new size, or limit in frames
response :		

6.4.2.14. Deframer Buffer Output Clock

<enum> :	3	fallback clock
	4	satellite clock
	6	external clock
	7	external G703
	8	external HSSI
request :	RIc?	get present status
	RIc<enum>	change status
response :	RIc<enum>	present or new status

6.4.2.15. Interface for Sub-D Connector

Remark that the Sub-D interface is at sub-address 1.

```

<enum> :      0      DVB-SPI-LVDS
               2      RS422-serial
               c      DVB-SPI-ECL
               d      HSSI
               f      Extended HSSI
               z      disabled

request :      :1RIs?          get present status
               :1RIs<enum>    change status

response :      :1RIs<enum>    send present or new status

```

6.4.2.16. Interface for Coaxial Connector

Remark that the Coaxial interface is at sub-address 2.

```

<enum> :      1      ECL (serial)
               3      electrical ASI
               5      optical ASI
               6      G703 - 8 Mbit/s
               7      G703 - 17 Mbit/s
               8      G703 - 34 Mbit/s
               9      G703 - 2 Mbit/s
               a      G703 - 6 Mbit/s
               b      G703 - 45 Mbit/s
               e      SSI
               z      disabled

request :      :2RIs?          get present status
               :2RIs<enum>    change status

response :      :2RIs<enum>    send present or new status

```

6.4.2.17. Modulation Standard

```

<enum> :      0      NTC
               1      SKYPLEX - SCPC
               2      SKYPLEX - TDMA
               3      DVB

request :      SDc?          get present status
               SDc<enum>    change status

response :      SDc<enum>    present or new status

```

6.4.3. Alarm Status

The enumeration for all alarms, unless mentioned otherwise :

<enum> :	0	status O.K.
	1	alarm
	2	not applicable
	9	masked

6.4.3.1. Reset All Memorized Alarms

The jointly reset of all memorized alarms can be done via the "all alarm string" request (see 6.4.3.19.).

6.4.3.2. Incompatibility Indication

request :	Ssi?	get present incompatibility indication
response :	Ssi###	send present incompatibility indication (decimal representation)

6.4.3.3.-6. +5.0 Volt, -5.0 Volt, +15.0 Volt and -15.0 Volt Power Supply

request :	Sza?	get measured +5.0V power supply voltage
	Szb?	get measured -5.0V power supply voltage
	Szc?	get measured +15.0V power supply voltage
	Szd?	get measured -15.0V power supply voltage
	Ssv?	get present power supply alarm status (summary)
	SsV?	get memorized power supply alarm status (summary)
response :	Szas###	measured +5.0V power supply voltage in volt
	Szbs###	measured -5.0V power supply voltage in volt
	Szcs###	measured +15.0V power supply voltage in volt
	Szds###	measured -15.0V power supply voltage in volt
response :	Ssv<enum>	present power supply alarm status (summary)
	SsV<enum>	memorized power supply alarm status (summary)

Remark that the measurable limits are preceded by the "<"- or ">"-signs and that the **Ssv** message indicates alarm if at least one of the four power supplies is (or has been) out of range.

6.4.3.7. Device Temperature

request :	Stt?	get present alarm status
	StT?	get memorized alarm status
	StT!	get memorized alarm status with reset
response :	Stt<enum>	present alarm status
	StT<enum>	memorized alarm status

6.4.3.8. Synthesiser

request :	Rsp?	get present alarm status
	RsP?	get memorized alarm status
	RsP!	get memorized alarm status with reset
response :	Rsp<enum>	present alarm status
	RsP<enum>	memorized alarm status

6.4.3.9. Clock PLL

request :	Rdc?	get present alarm status
	RdC?	get memorized alarm status
	RdC!	get memorized alarm status with reset
response :	Rdc<enum>	present alarm status
	RdC<enum>	memorized alarm status

6.4.3.10. Input Level Saturation

request :	Rls?	get present alarm status
	RIS?	get memorized alarm status
	RIS!	get memorized alarm status with reset
response :	Rls<enum>	present alarm status
	RIS<enum>	memorized alarm status

6.4.3.11.-12. Buffer Overflow and Underflow

<enum> :	0	status O.K.
	1	alarm
	2	buffer overflow
	3	buffer underflow
	9	masked
request :	Rio?	get present alarm status
	RiO?	get memorized alarm status
	RiO!	get memorized alarm status with reset
response :	Rio<enum>	present alarm status
	RiO<enum>	memorized alarm status

6.4.3.13. Data Interface

request :	Ris?	get present alarm status
	RiS?	get memorized alarm status
	RiS!	get memorized alarm status with reset
response :	Ris<enum>	present alarm status
	RiS<enum>	memorized alarm status

6.4.3.14. Demodulator Sync

request :	Rdl?	get present status
	RdL?	get memorized status
	RdL!	get memorized status with reset
response :	Rdl<enum>	present status
	RdL<enum>	memorized status

6.4.3.15. Viterbi Decoder Sync

No remote M&C command.

6.4.3.16. Unique Word Detection

No remote M&C command.

6.4.3.17. Reed-Solomon Decoder Sync

request :	Rcl?	get present status
	RcL?	get memorized status
	RcL!	get memorized status with reset
response :	Rcl <enum>	present status
	RcL <enum>	memorized status

6.4.3.18. Deframer Sync

request :	Rel?	get present status
	ReL?	get memorized status
	ReL!	get memorized status with reset
response :	Rel <enum>	present status
	ReL <enum>	memorized status

6.4.3.19. All Alarms

alarms :	a : controller had reset	(Srs - see 6.4.1.14.)
	b : RF transmit status (on/off)	(n.a.)
	c : power on self test	(Sss - see 6.4.1.12.)
	d : power supply alarm	(Ssv - see 6.4.3.3-6.)
	e : device temperature	(Stt - see 6.4.3.7.)
	f : transmit synthesiser	(n.a.)
	g : transmit HPA	(n.a.)
	h : transmit summary alarm	(n.a.)
	i : (receive) synthesiser	(Rsp - see 6.4.3.8.)
	j : receive summary alarm	(logic OR of alarms p, r, s and t)
	k : 10.0 MHz oscillator	(n.a.)
	l : external reference for 10.0 MHz	(n.a.)
	m : RF phase lock DRO	(n.a.)
	n : RF-head communication	(n.a.)
	o : incompatibility indication (summary)	(1 = incomp, Ssi - see 6.4.3.2.)
	p : clock PLL	(Rdc - see 6.4.3.9.)
	q : external clock for buffer output	(n.a.)
request :	Saa?	get present alarms
	SaA?	get memorized alarms
	SaA!	get memorized alarms with reset
response :	Saa abcdefghijklmnopqrst	present alarms
	SaA abcdefghijklmnopqrst	memorized alarms

6.4.3.20. General Device Alarm

<enum> :	0	device OK
	1	device alarm (logical-OR of alarms a, c, d, e, l, j, o, p, r, s and t, see 6.4.3.19.)
request :	Ssa?	get present alarm status
	SsA?	get memorized alarm status
response :	Ssa <enum>	present alarm status
	SsA <enum>	memorized alarm status

6.4.3.21. Demodulator Ready or Receiver Ready

<enum> :	0	receiver not ready
	1	receiver ready - demodulator sync (see 5.3.14.), viterby decoder sync (see 5.3.15.), unique word detected (see 5.3.16.), RS-decoder sync (see 5.3.17) and deframer sync (see 5.3.18.)
request :	Rss?	get present status
response :	Rss <enum>	present status

6.4.3.22. Setting the Threshold for the Byte Error Rate before RS-decoder

Request :	RBt?	get present Ber Threshold setting
	RBt<	get maximum Ber Threshold setting
	RBt>	get minimum Ber Threshold setting
	RBt####	change Ber Threshold range as requested
Response :	RBt####	send present/min/max/new Ber Threshold setting

6.4.3.23. Enabling / Disabling the BER Alarm

<enum> = 0 BER Alarm is not used
 1 BER Alarm used

Request :	RBs?	get the present setting of the BER Alarm (ON or OFF)
Response:	RBs <enum>	send the present setting

6.4.3.24. Get or clear the Ber alarms

RbA and Rba

<enum> = 0 No Ber Alarm (Byte Error Rate before RS-decoder is below the Ber Threshold)
1 Ber Alarm (Byte Error Rate before RS-decoder is over the Ber Threshold)

Request : Rba? get present Ber Alarm status
RbA? get memorized Ber Alarm status
RbA! get memorized Ber Alarm status with reset Ber alarm counter

Response : Rba<enum> send present Ber Alarm status
RbA<enum> send memorized Ber Alarm status

6.4.4. Monitor Parameters

6.4.4.1. Internal Temperature

request : **Sst?** get temperature
response : **Ssts##.#** temperature in °C

6.4.4.2. Receive Level

request : **Rll?** get receive level
 Rlc? get Co-level density
response : **Rlls##.##** receive level in dBm
 Rlcs###.## Co-level density in dBm/Hz

6.4.4.3. Eb/No Estimation

request : **Rle?** get Eb/No estimation
response : **Rles##.##** Eb/No estimation in dB

6.4.4.4. Carrier Frequency Offset

request : **Rfd?** get carrier frequency offset
response : **Rfds#####** carrier frequency offset in Hz

6.4.4.5. Symbol Rate Offset

request : **Rrd?** get symbol rate offset
response : **Rrds###.##** symbol rate offset in Hz

6.4.4.6. Symbol Error Rate before FEC

request : **Rcb?** get BER before FEC
response : **Rcb#####E-6** BER before FEC in E-6

6.4.4.7. Byte Error Rate before RS-decoder

request : **Rlb?** get byte error rate after Trellis Decoder
 response : **Rlb#####E-6** byte error rate after Trellis Decoder in E-6

6.4.4.8. Bit Error Rate after decoding

request : **Rub?** get BER after decoding
 response : **Rub#.##E-##** estimated BER after decoding

6.4.4.9. Link Margin

request : **Rld?** get link margin
 response : **Rlds###.##** estimated link margin in dB

6.4.4.10. PRBS Bit Error Rate

request : **Rpb?** get PRBS bit error rate
 response : **Rpb#####** PRBS bit error rate in errors per second

6.4.4.11. PRBS Bit Error Count

engineering string 000 read counters
 999 reset counters

request : **\$###** get engineering string
\$<text> engineering string

6.4.4.12. Buffer Contents

request : **Rib?** get buffer content in bytes
Rif? get buffer content in frames

response : **Rib#####** buffer content in bytes
Rif###.## buffer content in frames

6.4.4.13. Internal Test Variables (Test Mode Only)

Not applicable via remote interface. Reserved for in factory test and problem solving.

6.4.5. Test Menu

Not applicable via remote interface. Reserved for in factory test and problem solving.

6.4.6. Error Messages

<error code> :	000	undefined error
	020	undefined header error
	021	header missing or incomplete
	022	header contains invalid character(s)
	023	header unknown
	030	undefined data error
	031	data missing or incomplete
	032	data contains invalid character
	033	data length error
	034	data syntax error
	110	argument not acceptable (not further detailed)
	111	argument out of the acceptable range (enumeration)
	112	argument too low (numeric values)
	113	argument too high (numeric values)
request :	any valid message, which can not be handled	
response :	<NAK><address> err <error code><ETX><LRC>	

7. SERVICE

CAUTION : Servicing for qualified personnel only

Warning : The opening of the cover is likely to expose dangerous voltages. Disconnect the instrument from the voltage source before opening it :

- A. Switch off the power switch on the instrument's rear panel
- B. Disconnect the power supply plug
- C. Remove possible other signal carrying cables

7.1. Maintenance

There is no particular maintenance requirement for this device.

Procedure for opening the device :

- take care that the device is powered-down and that the external cables are disconnected
- remove the 26 screws from the cover and lift off the cover

Procedure for closing the device :

- place the cover and fasten the 26 cover screws
- reconnect the device

7.2. Fan Replacement

There are two fans located at the back panel of the device. These are low voltage DC fans from Multicomp, delivered by Farnell as part number 926 127. They have a life expectancy of more than 50.000 hours, this is more than 5 years, at 40°C.

Technical characteristics :
12 Vdc
1 W
40 x 40 x 20 mm
3.4 litres/sec air flow.

They can be ordered through Newtec Cy.

In the case of failure, take the following steps :

- open the device (see 7.1.)
- disconnect the fan cables on the baseband board

- remove the four fan screws, spacers and nuts
- lift the fan away from the rear panel
- lift the fan guard away from the rear panel
- install the new fan, take care for the correct orientation : the arrow pointing to the rear panel, as seen from inside and the two wires at the bottom side
- connect the fan cables to the connector on the baseband board, take care for the correct polarity : the blue wire at “-“ and the red wire at “+”
- close the device (see 7.1.)

7.3. Line Fuse Repair

The recommended fuses are size 5 by 20 mm, rate T 3.15 A, 250 V (UL recognised and IEC approved).

Replace only with same type and rating. The use of other fuses or material is prohibited.

The line fuses are housed in a small container above the rearpanel power supply connectors. This container is attached to the line module ; it cannot be removed.

To check the fuses, insert the tip of a screwdriver in the slot at the middle of the container and pry gently to extend the container.

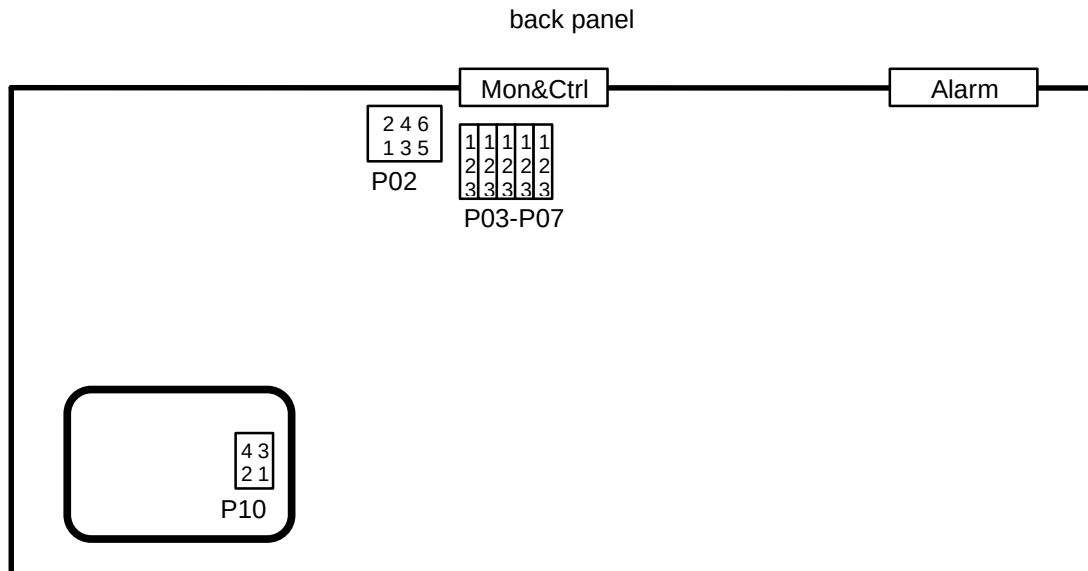
7.4. Installation Interface Module NTC/3xxx/xx

After the replacement of the interface board, the unit automatically recognises it and adapts its operation accordingly.

The procedure for opening the device is given in section 7.1.

7.5. Strap Positions

Straps on baseband board NTC/3306/AA :



The remote interface selection RS485 - RS232 is done via straps P03-P07 :

- position 1-2 for RS485
- position 2-3 for RS232

The RS485 are default selected (position 1-2) when the device leaves factory, unless mentioned otherwise via an indication on the back panel.

The procedure for opening the device, when these straps needs to be moved, is given in section 7.1..

The other straps are set from factory at the following positions :

strap	position	
P02	3-4	shield to ground (strapped)
P10	1-2	fixed

7.6. System Test Failure

After power-on, the device performs a self test and shows the result on the start-up display.

A failed self test is indicated by an error number which provides information about the origin of the defect. This is an hexadecimal bitmap, which refers to the different tests (bit 7 = MSB) :

- bit 7 : permanent memory error (system)
- bit 6 : permanent memory error (configuration)
- bit 5 : algorithm loopback error
- bit 4 : decoder loopback error
- bit 3 : viterbi decoder loopback error
- bit 2 : RS decoder loopback error
- bit 1 : demodulator loopback error
- bit 0 : interface loopback error

- A “**permanent memory error**” indicates that an error is detected in the LRC, which protect the EEPROM of the device. This may indicate that one of the settings is erroneous.

To solve this problem, the operator has to verify all the settings of the device in the system and/or control menu.

The correction of the possible wrong setting(s) in the system menu writes them immediately to EEPROM and recalculates the LRC.

The error in a configuration is solved by correcting it and then saving it again (see 5.1.2.). Each of the saved configurations has to be verified and corrected if necessary.

If the error(s) persists or reappears, then the device must be returned for repair.

- A “**loopback error**” indicates that the considered part has not correctly responded to the communication with the device controller. It is suggested to power-off and again -on the device. If the error(s) persists or reappears, then a device revision is required.

8. TROUBLESHOOTING REPORT

In order to get a quick response for the settlement of problems requiring assistance from NEWTEC Cy, the information sheet hereafter should be copied, filled in and sent back to NEWTEC Cy, Laarstraat 5, B-9100 Sint Niklaas (Belgium) - Fax : +32 3 780 65 49.

It is recommended to complete the troubleshooting report with information on the surrounding equipment and the related interfaces. This contains among other things a block diagram of the whole system, with type indications and connections, just as troubleshooting reports of these devices. Also logging files, from RMCP or any other remote M&C equipment (if available) covering the considered problem, should be included.

CUSTOMER :		Fax :
Equipment Type Number : NTC/2063/...		
Serial Number (backpanel) : SN...		
Date :		
INITIAL DISPLAY AFTER POWER ON :		
High Rate DVB+8+16 Demodulator system test ...		v... passed - failed ...
5.1. System Menu		
5.1.1. Load Configuration	N.A.	
5.1.2. Save Configuration	N.A.	
5.1.3. Test Functions	enabled - disabled	
5.1.4. Alarm Mask	hexadecimal number :	
5.1.5. Device Address for Serial Interface	number :	
5.1.6. Device Control Mode	local - remote	
5.1.7. Device Identification – hardware firmware	NTC/2063/... NTC/6094/AA v...	
5.1.8. Device Serial Number	number : SN...	
5.1.9. Device Capability	hexadecimal number :	
5.1.10. Interface Identification	NTC/3...	
5.1.11. LNB Power Supply Control	input A : enabled - disabled input B : enabled - disabled	
5.2. Control Menu		
5.2.1. Interface Rate	Mbit/s :	
5.2.2. Symbol Rate	Mbaud :	
5.2.3. Modulation FEC-rate	QPSK - 8PSK - 16QAM 1/2 - 2/3 - 3/4 - 5/6 - 7/8 - 8/9	
5.2.4. LNB A Control	N.A. – horizontal - vertical – high - low	
5.2.5. LNB B Control	N.A. – horizontal - vertical – high - low	
5.2.6. LNB Input	A - B – L-band - IF	
5.2.7. Excess Bandwidth	auto - 25% - 35%	
5.2.8. IF(L) Frequency	MHz :	
5.2.9. Carrier Acquisition Range	kHzptp :	
5.2.10. Spectrum Inversion	auto - off - on	
5.2.11. Descrambler	on - off	
5.2.12. MPEG Framing	external 188 - external 204 - internal	
5.2.13. Deframer Buffer Size	frames :	
5.2.14. Deframer Buffer Output Clock	satellite - external - internal	
5.2.15. Interface for Sub-D Connector	off - SPI - RS422- HSSI - ...	
5.2.16. Interface for Coaxial Connector	off - el. - opt. ASI - ECL - G.703 ... MHz - ...	
5.2.17. Modulation Standard	DVB - NTC - SKYPLEX SCPC - SKYPLEX TDMA	

5.3. Alarm Menu	
5.3.1. Reset All Memorized Alarms	N.A.
5.3.2. Incompatibility Indication	O.K. - incompatibility (hex.numb.) :
5.3.3. +5.0 Volt Power Supply	v : ... - O.K. - AL. - mem.al. - N.A. + number :
5.3.4. -5.0 Volt Power Supply	v : ... - O.K. - AL. - mem.al. - N.A. + number :
5.3.5. +15.0 Volt Power Supply	v : ... - O.K. - AL. - mem.al. - N.A. + number :
5.3.6. -15.0 Volt Power Supply	v : ... - O.K. - AL. - mem.al. - N.A. + number :
5.3.7. Device Temperature	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.8. Synthesiser	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.9. Clock PLL	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.10. Input Level Saturation	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.11. Buffer Overflow	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.12. Buffer Underflow	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.13. Data Interface	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.14. Demodulator Sync	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.15. Viterbi Decoder Sync	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.16. Unique Word Detection	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.17. Reed-Solomon Decoder Sync	O.K. - ALARM - mem.alarm - N.A. + number :
5.3.18. Deframer Sync	O.K. - ALARM - mem.alarm - N.A. + number :
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5.4.8. Bit Error Rate after decoding	...
5.4.9. Link Margin	dB :
5.4.10. PRBS Bit Error Rate	errors/sec :
5.4.11. PRBS Bit Error Count	N.A. - differences : ... - seconds :
5.4.12. Buffer Contents	frames :
5.5. Test Menu	
Front panel LED's	
Demodulator	green - off
Decoder	green - off
Deframer	green - off
Active Alarm	red - off
Memorized Alarm	red - off
Test	yellow - off
Additional comments and remarks : (possible causes for failure, missing signals, components, ...)	

Please copy this form and complete every input, return to :
NEWTEC Cy, Laarstraat 5, B-9100 Sint Niklaas (Belgium)
Tel : +32 3 780 65 00 - Fax : +32 3 780 65 49 - E-mail : techsupport@newtec.be

APPENDIX : APPLICATION NOTES

- NTC/2063/APN01 : BER Measurement Method for QPSK/8PSK/16QAM
- NTC/2063/APN02 : BER Performance DVB+8+16 Demodulator
- NTC/2062/APN05 : Channel Equalisation for High Baud Rate Modulators

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